

Uncertainty

CHAPTER 14

Outline

- ◇ Uncertainty
- ◇ Probability
- ◇ Syntax
- ◇ Semantics
- ◇ Inference rules

Uncertainty

Let action A_t = leave for airport t minutes before flight
Will A_t get me there on time?

Problems:

- 1) partial observability (road state, other drivers' plans, etc.)
- 2) noisy sensors (KCBS traffic reports)
- 3) uncertainty in action outcomes (flat tire, etc.)
- 4) immense complexity of modelling and predicting traffic

Hence a purely logical approach either

- 1) risks falsehood: " A_{25} will get me there on time"
- or 2) leads to conclusions that are too weak for decision making:
" A_{25} will get me there on time if there's no accident on the bridge
and it doesn't rain and my tires remain intact etc etc."

(A_{1440} might reasonably be said to get me there on time
but I'd have to stay overnight in the airport ...)

Methods for handling uncertainty

Default or nonmonotonic logic:

Assume my car does not have a flat tire

Assume A_{25} works unless contradicted by evidence

Issues: What assumptions are reasonable? How to handle contradiction?

Rules with fudge factors:

$A_{25} \mapsto_{0.3} \text{get there on time}$

$\text{Sprinkler} \mapsto_{0.99} \text{WetGrass}$

$\text{WetGrass} \mapsto_{0.7} \text{Rain}$

Issues: Problems with combination, e.g., *Sprinkler* causes *Rain*??

Probability

Given the available evidence,

A_{25} will get me there on time with probability 0.04

Mahaviracarya (9th C.), Cardano (1565) theory of gambling

(Fuzzy logic handles *degree of truth* NOT uncertainty e.g.,
 WetGrass is true to degree 0.2)

Probability

Probabilistic assertions *summarize* effects of

laziness: failure to enumerate exceptions, qualifications, etc.

ignorance: lack of relevant facts, initial conditions, etc.

Subjective or Bayesian probability:

Probabilities relate propositions to one's own state of knowledge

e.g., $P(A_{25} | \text{no reported accidents}) = 0.06$

These are not assertions about the world

Probabilities of propositions change with new evidence:

e.g., $P(A_{25} | \text{no reported accidents, 5 a.m.}) = 0.15$

(Analogous to logical entailment status $KB \models \alpha$, not truth.)

Making decisions under uncertainty

Suppose I believe the following:

$$P(A_{25} \text{ gets me there on time} | \dots) = 0.04$$

$$P(A_{90} \text{ gets me there on time} | \dots) = 0.70$$

$$P(A_{120} \text{ gets me there on time} | \dots) = 0.95$$

$$P(A_{1440} \text{ gets me there on time} | \dots) = 0.9999$$

Which action to choose?

Depends on my preferences for missing flight vs. airport cuisine, etc.

Utility theory is used to represent and infer preferences

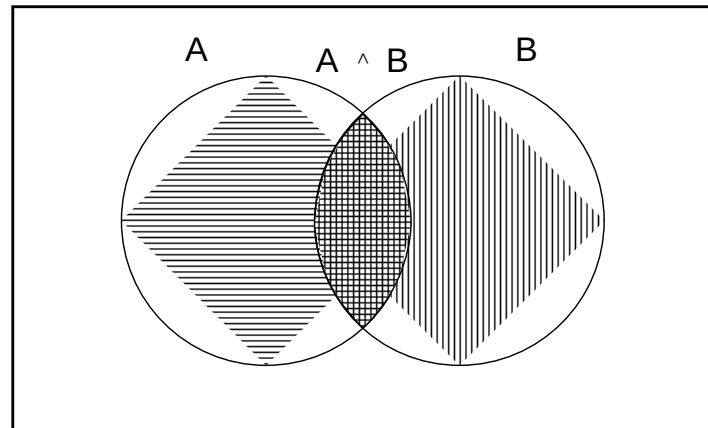
Decision theory = utility theory + probability theory

Axioms of probability

For any propositions A , B

1. $0 \leq P(A) \leq 1$
2. $P(\text{True}) = 1$ and $P(\text{False}) = 0$
3. $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

True



de Finetti (1931): an agent who bets according to probabilities that violate these axioms can be forced to bet so as to lose money regardless of outcome.

Syntax

Similar to propositional logic: possible worlds defined by assignment of values to random variables.

Propositional or Boolean random variables

e.g., *Cavity* (do I have a cavity?)

Include propositional logic expressions

e.g., $\neg \textit{Burglary} \vee \textit{Earthquake}$

Multivalued random variables

e.g., *Weather* is one of $\langle \textit{sunny}, \textit{rain}, \textit{cloudy}, \textit{snow} \rangle$

Values must be exhaustive and mutually exclusive

Proposition constructed by assignment of a value:

e.g., $\textit{Weather} = \textit{sunny}$; also $\textit{Cavity} = \textit{true}$ for clarity

Syntax contd.

Prior or unconditional probabilities of propositions

e.g., $P(Cavity) = 0.1$ and $P(Weather = sunny) = 0.72$
correspond to belief prior to arrival of any (new) evidence

Probability distribution gives values for all possible assignments:

$P(Weather) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)

Joint probability distribution for a set of variables gives
values for each possible assignment to all the variables

$P(Weather, Cavity) =$ a 4×2 matrix of values:

$Weather =$	$sunny$	$rain$	$cloudy$	$snow$
$Cavity = true$				
$Cavity = false$				

Syntax contd.

Conditional or posterior probabilities

e.g., $P(Cavity|Toothache) = 0.8$

i.e., given that *Toothache* is all I know

Notation for conditional distributions:

$\mathbf{P}(Weather|Earthquake)$ = 2-element vector of 4-element vectors

If we know more, e.g., *Cavity* is also given, then we have

$P(Cavity|Toothache, Cavity) = 1$

Note: the less specific belief *remains valid* after more evidence arrives, but is not always *useful*

New evidence may be irrelevant, allowing simplification, e.g.,

$P(Cavity|Toothache, 49ersWin) = P(Cavity|Toothache) = 0.8$

This kind of inference, sanctioned by domain knowledge, is crucial

Conditional probability

Definition of conditional probability:

$$P(A|B) = \frac{P(A \wedge B)}{P(B)} \text{ if } P(B) \neq 0$$

Product rule gives an alternative formulation:

$$P(A \wedge B) = P(A|B)P(B) = P(B|A)P(A)$$

A general version holds for whole distributions, e.g.,

$$\mathbf{P}(Weather, Cavity) = \mathbf{P}(Weather|Cavity)\mathbf{P}(Cavity)$$

(View as a 4×2 set of equations, *not* matrix mult.)

Chain rule is derived by successive application of product rule:

$$\begin{aligned} \mathbf{P}(X_1, \dots, X_n) &= \mathbf{P}(X_1, \dots, X_{n-1}) \mathbf{P}(X_n|X_1, \dots, X_{n-1}) \\ &= \mathbf{P}(X_1, \dots, X_{n-2}) \mathbf{P}(X_{n-1}|X_1, \dots, X_{n-2}) \mathbf{P}(X_n|X_1, \dots, X_{n-1}) \\ &= \dots \\ &= \prod_{i=1}^n \mathbf{P}(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$